

FELIX NWOKEKEH

Greater Noida, India | +91 8826357820 | fnwokekeh@gmail.com

Professional Summary

AI & Machine Learning Engineer | Data Scientist

Passionate about building **AI-powered solutions** in healthcare, research, and business. Skilled in **deep learning, predictive modeling, and generative AI**, with hands-on experience in developing real-world applications such as MRI diagnostic systems and multimodal RAG assistants. Certified by **Google, Stanford, and IBM**, with a strong foundation in data-driven problem solving and innovation.

Technical Skills

- **Programming:** Python, R, SQL (MySQL), Java, C, HTML, CSS
- **ML & AI:** TensorFlow, scikit-learn, CNNs, LSTMs, OpenCV, LangChain, Deep Learning
- **Data Science Tools:** Jupyter Notebook, Google Colab, Spyder, Excel, Tableau
- **Core Expertise:** Predictive Modeling, Computer Vision, Generative AI, Data Visualization
- **Soft Skills:** Communication, Collaboration, Problem-Solving, Time Management

Projects

MedMRI: AI-Powered MRI Analysis Web App

- Built an end-to-end MRI analysis application using **Google Gemini AI & Streamlit**.
- Performed automated detection of abnormalities across **6+ MRI types** (brain, knee, spleen, pelvis, spine, heart).
- Achieved **95% accuracy** in abnormality detection and generated reports in **under 10 seconds per scan**.
- Integrated **user authentication + relational DB storage** with automated **PDF reporting**, reducing reporting time by **70%**.

PDF Q&A Application (RAG with Llama-2 & LangChain)

- Developed a **Retrieval-Augmented Generation system** to query and summarize PDF files.
- Reduced information retrieval time by **80%** compared to manual search.
- Achieved **>85% semantic relevance** in benchmark testing with research documents.

Brain Tumor Detection (CNN)

- Implemented a CNN pipeline for tumor detection from MRI scans.
- Achieved **91% classification accuracy** across multiple test datasets.

- Reduced diagnostic error rate by **18%** compared to baseline models.

Heart Failure Prediction (ML Classification)

- Built a model using logistic regression & ensemble methods.
- Improved F1-score to **0.87**, enabling early risk detection with **83% sensitivity**.
- Reduced false negatives by **15%** via optimized feature engineering.

Car Price Prediction (Regression)

- Created regression-based predictive model on 10k+ car listings.
- Achieved **R² score of 0.89** and **RMSE of 2,800** on test dataset.

House Price Prediction (Regression)

- Developed regression model on housing datasets with 20+ features.
- Achieved **R² score of 0.91** and **RMSE of 3,200**, outperforming baseline models by **12%**.

Dogs vs Cats Image Classification (CNN)

- Built CNN achieving **96% test accuracy** on Kaggle dataset with real-time prediction support.

Emotion Recognition (OpenCV + CNN)

- Real-time facial emotion detection with **87% accuracy** on test data.
- Successfully classified 6+ emotions in **under 2 seconds per frame**.

Data Analysis & Visualization Projects

- **Amazon Sales Dashboard (Tableau):** Analyzed 100k+ sales records; increased KPI visibility by **40%**.
- **Financial Complaints Dashboard:** Tracked resolution times; reduced issue-tracking lag by **30%**.
- **COVID-19 Dataset EDA:** Processed 1M+ records for trend forecasting and case growth modeling.

Certifications & Achievements

- **Google Data Analytics Certification** – Data collection, analysis & visualization
- **Machine Learning Specialization** – Andrew Ng, Stanford University
- **IBM Data Science Professional Certificate**
- Recognized for **outstanding performance** in Data Science Bootcamp

Interests & Languages

- Interests: Artificial Intelligence, Computer Vision, Prompt Engineering, Music, Football, Reading
- Languages: English (Fluent), French (Intermediate)